Automated Fertilizer Granules Sprayer Using Raspberry Pi Pico

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Project Type: Mini Project / Prototype

1. Introduction

This project titled "Automated Fertilizer Granules Sprayer using Raspberry Pi Pico" is a mini project developed as a prototype to demonstrate the concept of automated and controlled fertilizer application in agriculture. The main objective is to reduce manual effort and fertilizer wastage by using sensors and a microcontroller-based system.

2. Problem Statement

Traditional fertilizer application methods are mostly manual, leading to uneven distribution, excess usage, and increased labor. There is a need for an automated system that can apply fertilizer only when required and at specific locations. Mainly my father face this as an issue due to lack of labor.

3. Project Description

This project is designed as a small working model that uses ultrasonic sensors to detect plants or obstacles and a servo motor to control the flow of fertilizer granules. The entire system is controlled using a Raspberry Pi Pico board programmed in MicroPython.

4. Components Used

- Raspberry Pi Pico

- Two Ultrasonic Sensors (HC-SR04)

- Servo Motor (SG90)

- DC Motors and Wheels

- Motor Driver Module

- Power Bank (for Pico board supply)

- Plastic Bottle for Fertilizer Storage

- Granules (Sand used for demonstration)

- Chassis and Connecting Wires

5. Working Principle

The ultrasonic sensors continuously measure the distance of objects in front of the vehicle. When a plant or obstacle is detected within a predefined distance, the Raspberry Pi Pico processes this data and activates the servo motor. The servo motor opens the outlet pipe, allowing fertilizer granules to fall. After a short delay, the servo motor returns to its original position, stopping the flow of granules.

6. Vehicle Movement Method

Due to the additional weight of the fertilizer granules, storage box, motors, and Pico board, the vehicle requires more power to move automatically. Providing sufficient battery power would require multiple batteries, which was not feasible due to cost constraints.

Therefore, in this prototype, a stick is attached to the vehicle and used to move it manually. This allows proper demonstration of the sensing and fertilizer spraying mechanism.

7. Power Supply Limitation

The Raspberry Pi Pico is powered using a power bank. Although this is sufficient for sensor and servo motor operation, it is not enough to drive the vehicle automatically under heavy load. Sand is used instead of real fertilizer like urea to safely demonstrate the system.

8. Applications

- Precision agriculture
- Automated gardening systems
- Small-scale farming
- Educational and research projects

9. Future Scope

This mini project is developed as a prototype. In the future, it can be converted into a real-time project by using:

- High-capacity rechargeable batteries
- Stronger motors with gear mechanisms
- GPS or line-following navigation
- Fertilizer level sensing and control
- Fully autonomous movement for real agricultural fields

10. Conclusion

This project successfully demonstrates the concept of an automated fertilizer granules sprayer using Raspberry Pi Pico, ultrasonic sensors, and a servo motor. Even with power limitations, the core idea and working mechanism are implemented successfully. This mini project serves as a foundation for developing a full-scale real-time agricultural solution.